

Verslog

Complete the table with equivalents Binary, octal and Hexa for 143-75 (decimal), 11011101 (binary), 453 (octal) 1F-C (hex)

Decimal	Binary	octal	Hexa
143-75	10001111-11	217-6	812-C
27-625	110111-101	33-5	1B-A
299	100101011	45-3	12B
31-75	11111-11	37-6	1F-C

now 143-75(10) $\rightarrow$ (2)(2)

2	143	
2	71	-1
2	35	-1
2	17	-1
2	8	-1
2	4	-0
2	2	

= 10001111

143-75(10) = 10001111-11 (2) //

now, $143 - 75(10) \rightarrow (2)(8)$

$$\begin{array}{r} 8 \overline{) 143} \\ \underline{8} \\ 17 \\ \underline{8} \\ 9 \end{array}$$

$\rightarrow 217$

$$\therefore 143 - 75(10) \rightarrow 217 - 6(8) //$$

now $143 - 75(10) \rightarrow (2)(16)$

$$\begin{array}{r} 16 \overline{) 143} \\ \underline{8} \\ 8 \end{array}$$

$$0 - 75 \times 16 = 120$$

$$\therefore 143 - 75(10) = 8B - 6 //$$

now $11011 - 101(2) \rightarrow (2)(10)$

$$1 \times 2^4 + 1 \times 2^3 + 0 + 1 \times 2 + 1 \times 2^0 + 1 \times 2^{-1} + 0 + 1 \times 2^{-3} = 27.625_{(10)}$$

now $11011 - 101 \rightarrow (3)(8) //$

$$\boxed{11011} \cdot 101 = 33 - 5(8) //$$

now $\boxed{11011} \cdot 1010 = 1B - 4(16) //$

$$\text{new } 453(8) \rightarrow (7)(10)$$

$$4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0 = 299(10)$$

$$\text{new } 453(8) \rightarrow (7)(2)$$

$$\begin{array}{|c|c|c|} \hline 100 & 101 & 011 \\ \hline 4 & 5 & \\ \hline \end{array} \rightarrow 100101011(2)$$

$$\text{new } 453(8) \rightarrow (7)(6)$$

$$\begin{array}{|c|c|c|} \hline 0001 & 0010 & 011 \\ \hline 1 & 2 & 11 \\ \hline \end{array} = 12B$$

Q3. Find the radix r of the following.

$$\frac{1}{r} (312/20)_r = (13-1)r$$

$$\Rightarrow \frac{312r}{20r} = 13-1r$$

$$\Rightarrow \frac{3 \times r^2 + 1 \times r + 2}{2 \times r + 0} = 1 \times r + 3 + 1 \times r^{-1}$$

$$\Rightarrow \frac{3r^2 + r + 2}{2r} = r + 3 + \frac{1}{r}$$

$$\Rightarrow 3r^2 + r + 2 = 2r^2 + 6r + 2$$

$$\Rightarrow r^2 - 5r = 0$$

$$\Rightarrow r^2 = 5r$$

$$\Rightarrow \boxed{r = 5}$$

$$2). 264r + 136r = 433r$$

$$\Rightarrow 2 \times r^2 + 6 \times r + 4 + 1 \times r^2 + 3 \times r + 6 = 4 \times r^2 + 3r + 3$$

$$\Rightarrow 2r^2 + 6r + 4 + r^2 + 3r + 6 = 4r^2 + 3r + 3$$

$$\Rightarrow 3r^2 + 9r + 10 = 4r^2 + 3r + 3$$

$$\Rightarrow -r^2 + 6r + 7 = 0$$

$$r = -1 \quad r = 7$$

r cannot be negative.

$$\text{So, } \boxed{r = 7}$$

$$3). \begin{matrix} 5 & 4 & 3 & 2 & 1 & 0 \\ 2 & 2 & 1 & 5 & 0 & 5 \end{matrix} (8)_2 \quad \begin{matrix} 4 & 3 & 2 & 1 & 0 \\ 1 & 2 & 3 & 4 & 5 \end{matrix} r$$

$$2 \times 8^5 + 2 \times 8^4 + 1 \times 8^3 + 5 \times 8^2 + 0 + 5 \times 1 = 1 \times 8^4 + 2 \times 8^3 + 3 \times 8^2 + 4 \times 8 + 5$$

$$\therefore \boxed{r = 16}$$

Q4) Addition

$\Rightarrow 1101(2) + 11011111(2)$

$$\begin{array}{r} 1101 \\ 11011111 \\ \hline 11101100 \end{array}$$

$\Rightarrow [11101100(2)]$

Q5) $77652(8) + 76543(8)$

$$\begin{array}{r} 77652 \\ 76543 \\ \hline 176415 \end{array}$$

$5+6=11(10) \Rightarrow 8 \overline{) 11} \begin{array}{l} 1 \\ -8 \\ \hline 3 \end{array} \uparrow 114$

$6+6=12(10)$

$28 \overline{) 12} \begin{array}{l} 1 \\ -8 \\ \hline 4 \end{array} \uparrow 144$

$[176415(8)]$

$7+7=14(10) \Rightarrow 8 \overline{) 14} \begin{array}{l} 1 \\ -8 \\ \hline 6 \end{array} \uparrow$

$8+7=15(10) \Rightarrow 8 \overline{) 15} \begin{array}{l} 1 \\ -8 \\ \hline 7 \end{array} \uparrow$

$$3) \quad 578A_{16} + ABDE_{16} \quad A+E = 24_{(10)} \quad 16 \overline{)24} \quad \underline{1-8} \quad 2^{16}_{(6)}$$

$$\begin{array}{r} 578A \\ + ABDE \\ \hline 10368 \end{array}$$

$$9+D = 22_{(10)} \quad 16 \overline{)22} \quad \underline{1-6} \quad 2^{16}_{(6)}$$

$$8+B = 19_{(10)} \quad 16 \overline{)19} \quad \underline{1-3} \quad 2^{16}_{(6)}$$

$$6+A = 16_{(10)} \quad 16 \overline{)16} \quad \underline{1-0} \quad 2^{16}_{(6)}$$

$$4) \quad 234_{(6)} + 315_{(6)}$$

$$\begin{array}{r} 234 \\ + 315 \\ \hline 553 \end{array}$$

$$= [553]_6$$

$$5+4 = 9_{(10)} \quad 6 \overline{)9} \quad \underline{1-3} \quad 2^{16}_{(6)}$$

$$3+2 = 5_{(10)} \quad 6 \overline{)5} \quad \underline{0-5} \quad 2^{05}_{(6)}$$

5) Binary $\rightarrow$ Gray Code (conversion)
(xor operation (bitwise xor))

3) $578A_{16} + ABDE_{16}$

$$\begin{array}{r} 578A \\ + ABDE \\ \hline 10368 \end{array}$$

$A+E = 24_{10}$ $16 \overline{) 24}$
 $1-8$

$9+0 = 9_{10}$ $16 \overline{) 9}$
 $1-5$

$8+B = 19_{10}$ $16 \overline{) 19}$
 $1-3$

$6+A = 16_{10}$ $16 \overline{) 16}$
 $1-0$

4) $234_{10} + 315_{10}$

$$\begin{array}{r} 234 \\ + 315 \\ \hline 553 \end{array}$$

$[553]_{10}$

$5+4 = 9_{10}$ $6 \overline{) 9}$
 $1-3$

$3+2 = 5_{10}$ $6 \overline{) 5}$
 $0-5$

5) Binary $\rightarrow$ Gray Code (conversion)
(xor operation bitwise xor)

$$\Rightarrow 10101(2)$$

$$^2 \begin{array}{c} 1 - 0 - 1 - 0 - 1(2) \\ \text{xor} \quad \text{xor} \quad \text{xor} \quad \text{xor} \quad \text{xor} \end{array}$$

$$^2 [11112]$$

$$\Rightarrow 01110(2)$$

$$^2 \begin{array}{c} 0 - 1 - 1 - 1 - 0(2) \\ \text{xor} \quad \text{xor} \quad \text{xor} \quad \text{xor} \quad \text{xor} \end{array}$$

$$^2 [01001(2)]$$

Q6. Gray code & Binary conversion

$$\Rightarrow 01101(\text{Gray})$$

$$\begin{array}{r} 01101(\text{gray}) \\ 01001(2) \\ \hline \hline \end{array}$$

$$^2 [01001(2)]$$

$$27. 1010 \text{ (gray)}$$

$$\begin{array}{cccc} 1 & 0 & 1 & 0 \text{ (gray)} \\ 1 & 1 & 0 & 0 \text{ (2)} \end{array}$$

$$\underline{\underline{[1100]_{(2)}}}$$

now, $1F - 6_{(10)} \rightarrow (3)_{(10)}$

$$1 \times 16^1 + 15 \times 16^0 + 12 \times 16^{-1} = 31.75_{(10)}$$

now, $1F - 6_{(16)} \rightarrow (2)_{(2)}$

$$> 00011111 - 100_{(2)}$$

now, $1F - 6_{(16)} \rightarrow (3)_{(8)}$

$$\underline{011}111 - 110 \rightarrow 37.6_{(8)}$$

28. Power equivalent of the following:

$$324_{(5)} = (3)_{(10)}$$

$$3 \times 5^2 + 2 \times 5^1 + 4^2 = [89]_{(10)}$$

$$2. \quad 32 - 23(4) = (2)_{10}$$

$$3 \times 4^1 + 2 + 2 \times 4^{-1} \times 3 \times 4^{-2} = \underline{\underline{[14.6875]_{(10)}}}$$

$$3. \quad 231_{(10)} = (?)_7$$

$$\begin{array}{r} 7 \overline{) 231} \\ \underline{33} \\ 7 \overline{) 33} \\ \underline{4} \\ 5 \end{array}$$

$$= [450]_{(7)}$$

$$4. \quad 189_{(10)} = (?)_6$$

$$\begin{array}{r} 6 \overline{) 189} \\ \underline{31} \\ 6 \overline{) 31} \\ \underline{5} \\ 1 \end{array}$$

$$= [513]_{(6)}$$